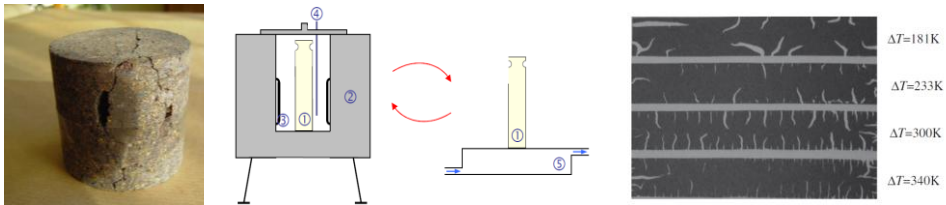


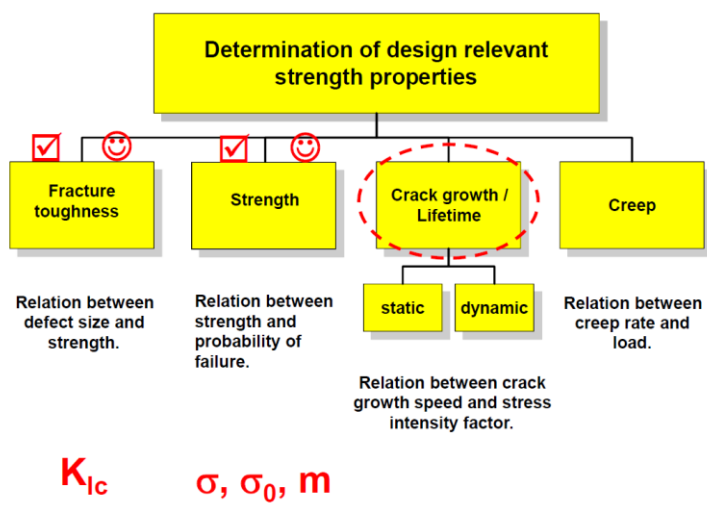
CERAMIC ENGINEERING

Chapter 4: Thermal and thermo mechanical properties

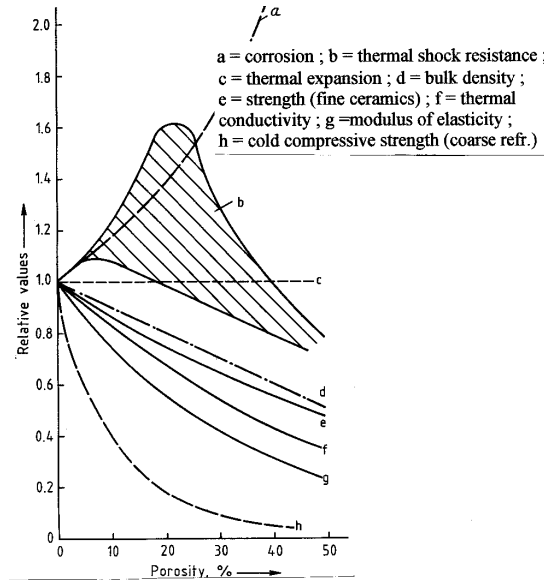


Courtesy of Chr. Aneziris and P. Gehre, Technical University of Freiberg, Germany

Chapter 4: Thermal and thermo mechanical properties



Chapter 4: Thermal and thermo mechanical properties



Ullmann's Encyclopedia of Industrial Chemistry,
Vol. A 23 (1993)

Ceramics and porosity

Influence of
porosity on
the properties
of ceramic
materials

3

Chapter 4: Thermal and thermo mechanical properties

Thermal expansion

Materials change their size when the temperature is changed:

$$\frac{l_1 - l_0}{l_0} = \alpha_{lin}(T_1 - T_0)$$

$$\alpha_{lin} = \frac{1}{l_0} \frac{dl}{dT}$$

where α_{lin} is the linear coefficient of thermal expansion, dT the temperature difference and dl the change in length.

Chapter 4: Thermal and thermo mechanical properties

Modes of heat transfer

There are three modes of heat transfer:

- **Conduction**

- Conduction is the transfer of heat through materials by the direct contact of matter.
- A diffusive process wherein molecules transmit their kinetic energy to other molecules by colliding with them.

- **Convection**

- Convection is the transfer of heat by the motion of a fluid medium (liquid or gas).

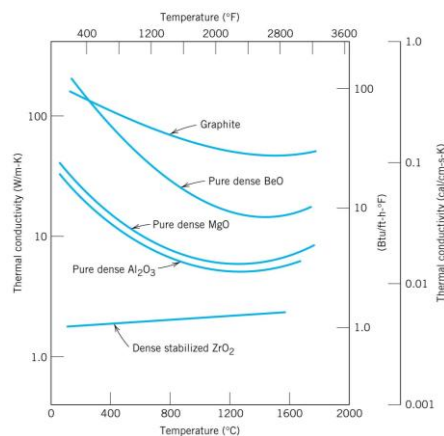
- **Radiation**

- The transfer of heat by electromagnetic waves (e.g. sunshine)

$$e_b(T) = \sigma T^4$$

Chapter 4: Thermal and thermo mechanical properties

Thermal conductivity



(From W. D. Kingery, H. K. Bowen, and D. R. Uhlmann, Introduction to Ceramics, 2nd ed., John Wiley & Sons, Inc., New York, 1976.)

Thermal conductivity of several ceramics over a range of temperatures

Chapter 4: Thermal and thermo mechanical properties

High Thermal Conductivity

Following factors promote high thermal conductivity:

- Low atomic mass
- Strong interatomic bonding (covalent in preference to ionic) for high Debye temperature
- Equivalently, high modulus of elasticity
- Simple crystal structure (less scattering of phonons)
- Low anharmonicity
- Isotopically pure

Examples:	Diamond	2000 (W/mK)
	SiC	490
	AlN	320

Rollet, A.D., De Graef, M.: 27-301 - Microstructure-Properties. Thermal Transport Properties. Lecture, Carnegie Mellon University, Pittsburgh, USA 2015.

Chapter 4: Thermal and thermo mechanical properties

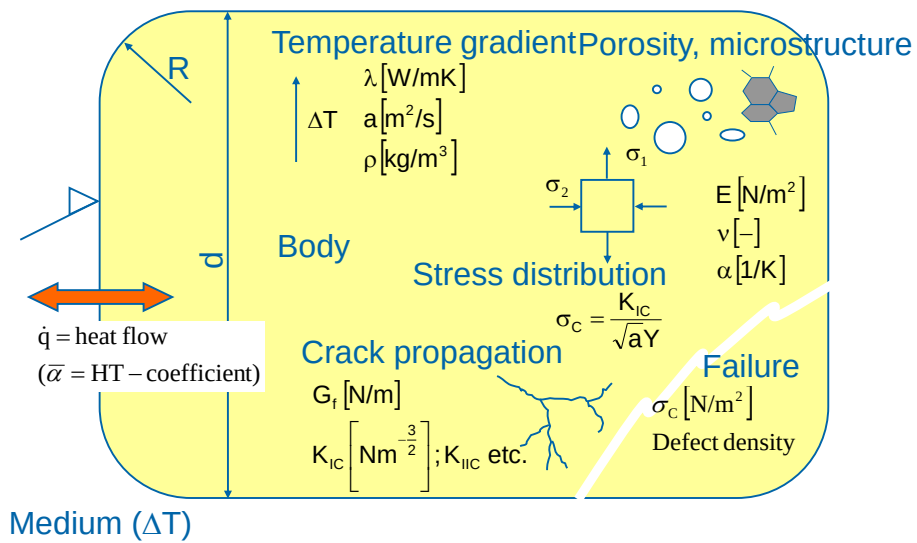
Low Thermal Conductivity

Following factors promote low thermal conductivity:

- Low density
- Low grain size (grain boundaries scatter phonons)
- High porosity
- Low pore size
- Any defects (including surfaces and interfaces) scatters phonons
- Complex crystal structure

Examples:	AAC	0.010 (W/mK)
	aerogels	0.005

Rollet, A.D., De Graef, M.: 27-301 - Microstructure-Properties. Thermal Transport Properties. Lecture, Carnegie Mellon University, Pittsburgh, USA 2015.



Source: Franken, Diss., 2000

Strength behaviour on thermal cycling

The TS characterizes the behaviour of a material or workpiece when exposed to the repeated influence of thermal stresses in a changing temperature field.

This behaviour is influenced by material microstructure, shape and volume of the loaded products as well as the external load conditions.

On heating and cooling of refractory materials, different stresses are caused as a result of mechanical stresses in the material.

Ceramic materials are exposed to particular danger on rapid cooling.

In this case, there is temperature gradient from the surface to the inside of the material in the form that the cooler material layers are on the outside. Consequently the outer material layers want to contract more and therefore become subject to relatively high tensile stresses.

11

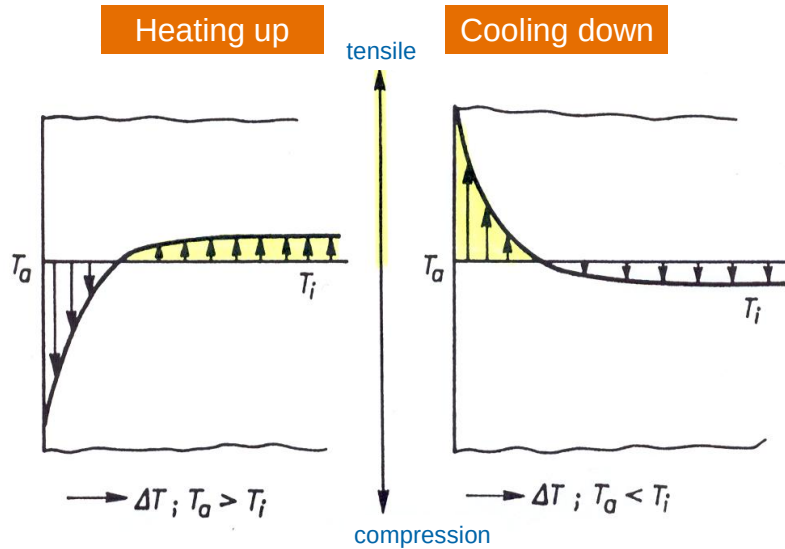
On rapid preheating, conditions are such that the outer material layers have Higher temperatures than the interior of the material and are therefore subject to compressive stresses!

The tensile stressess occurring simultaneously in the interior of the material Remain fairly low because they are distributed over a relatively wide layer of material.

Destruction caused by excessive compressive stress in the outer layers is caused mostly by the action of shear stresses, which are generated in their maximum extent at an angle of 45° to the effective compressive stress.

⇒spalling.

12



13

Key influencing parameters:

- Physical material characteristics
(strength, thermal expansion, thermal conductivity)
- Microstructure
(micro porosity, micro cracks, feather-like or hard dispersion phases)
- External loading conditions
(heat transfer conditions, temperature interval, sequence of T change, etc.)
- Shape of the loaded body according to size and degree of complexity.

14

Chapter 4: Thermal and thermo mechanical properties

Thermal stresses occur due to

- Restrained thermal expansion/contraction
- Temperature gradients that lead to differential dimensional changes

= => several layers like an onion!

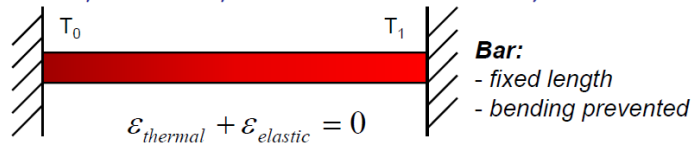


$$\sigma = E \alpha (T_0 - T_1) = E \alpha \Delta T$$

15

Chapter 4: Thermal and thermo mechanical properties

Stress will develop when component is fixed dimensionally.



$$\sigma_{thermal} = |E \cdot \varepsilon_{thermal}| = |E \cdot \varepsilon_{elastic}| = |E \cdot \alpha \cdot (T_1 - T_0)|$$

if $\varepsilon_{thermal} > \varepsilon_{elastic}$ → failure

16

Chapter 4: Thermal and thermo mechanical properties

A brass rod is stress free at room temperature.

It is heat but prevented from expanding in length

At what temperature T_ϕ does the stress reach -172MPa?

($E = 100 \text{ GPa}$ for brass, $\alpha = 20 \cdot 10^{-6} / ^\circ\text{C}$)

$$T_o - T_\phi = \sigma / E \alpha \Rightarrow$$

$$T_\phi = T_o - \sigma / E \alpha = >$$

$$T_\phi = 20 ^\circ\text{C} - (-1,72 \cdot 10^8 \text{Pa}) / ((10^{11} \text{Pa}) (20 \cdot 10^{-6} / ^\circ\text{C}))$$

$$T_\phi = 20 ^\circ\text{C} + 86 ^\circ\text{C} = 106 ^\circ\text{C}$$

17

Chapter 4: Thermal and thermo mechanical properties

Thermal stress fracture resistance
parameters

CRACK INITIATION

For an almost infinitely high heat transfer:

$$R_1 = \sigma (1-\mu) / \alpha E$$

BIOT number: $b = d k / \lambda$ with k heat transfer coefficient, d half the thickness of the body

For a finite heat transfer:

$$R_2 = \sigma (1-\mu) \lambda / \alpha E$$

18

Chapter 4: Thermal and thermo mechanical properties

Thermal stress fracture resistance
parameters

For an almost infinitely high heat transfer:

A third thermal stress parameter results when one looks at the maximum permissible cooling rate that a body can withstand without crack formation:

$R_3 = R_2 / (C_p \rho) = R_1 a$ where a is the thermal diffusivity!

→ Of great importance the strain at fracture: $\epsilon = \sigma / E$!

In the case of creep and flow phenomena R_{K1} and R_{K2} with η in the place of E .

$$a = \frac{\lambda}{\rho C_p}$$

19

Chapter 4: Thermal and thermo mechanical properties

Thermal stress damage resistance
parameters

CRACK PROPAGATION OR CRACK GROWTH

The crack formation discussed above does not necessarily lead to fracture, when crack growth proceeds slowly!

For instance for refractory materials (coarse grain materials), subcritical crack growth can be expected.

Resistance factor against crack propagation:

$$R_{St} = E \gamma N / (\sigma^2 (1-\mu))$$

with γ energy for new fracture surface formation, N number of cracks

20

CRACK PROPAGATION OR CRACK GROWTH

Hasselman considered a uniformly cooled or heated body, which contraction/expansion is fully restrained. Any temperature change ΔT of this body leads to a homogeneous thermal stress distribution σ . If $\sigma > \sigma_c$ (strength of the material), cracks were initiated. When fracture occurs, the ratio of total fracture energy G_t to stored elastic strain energy G_e should be therefore as high as possible. For a cube ($V = L^3$) holds at crack initiation

$$\frac{G_t}{G_e} = \frac{G_f L^2 E}{\sigma_c^2 (1 - \nu) V} = \frac{G_f E}{\sigma_c^2 (1 - \nu) L} = \frac{2 R''''}{L}$$

leading to the definition of the thermal stress damage resistance
par

$$R'''' = \frac{G_f E}{2 \sigma_c^2 (1 - \nu)}$$

Generally, R'''' has to be as large as possible to limit the kinetic cracks growth at crack initiation.

21

ASTM C 1421

“Standard Test Methods for Determination of Fracture Toughness of Advanced Ceramics at Ambient Temperature” is a high-quality, technically rigorous, full-consensus standard that may have finally answered the question,

“What is the ‘real’ fracture toughness of ceramics?”

Chapter 4: Thermal and thermo mechanical properties

The standard allows three tests methods: the chevron notch beam (CNB), the single edge-precracked-beam (SEPB), and the surface-crack in flexure (SCF).

Three test methods are allowed because no one method provides valid test results for the variety of ceramics being developed. All the test methods use 3- or 4-point flexure to test rectangular cross sections such as the common 3 mm by 4 mm test section used in flexural strength testing of ceramics.

More importantly, all have a well defined loading configuration, a well defined crack, and an accurate stress intensity factor coefficient solution.

Some allowances are made for other cross sections because adequate results can be obtained from such sections.

23

Chapter 4: Thermal and thermo mechanical properties

Despite the commonality of bend testing a rectangular cross section, the methods differ substantially in terms of the size, shape and development of the cracks used.

The SEPB uses a large, straight-through precrack that is rapidly formed from a small Vickers indentation or a shallow machined notch.

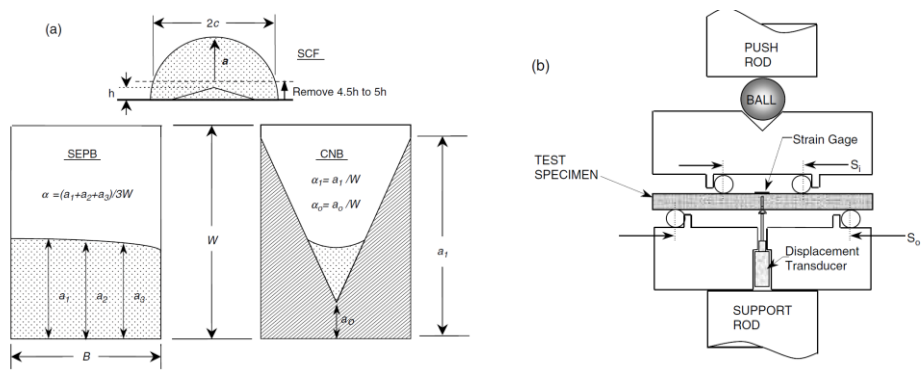
The CNB uses a 3-dimensional crack that is generated stably during fracture testing of the section.

The SCF uses a semi-elliptical precrack generated by removing a portion of a Knoop indentation precrack.

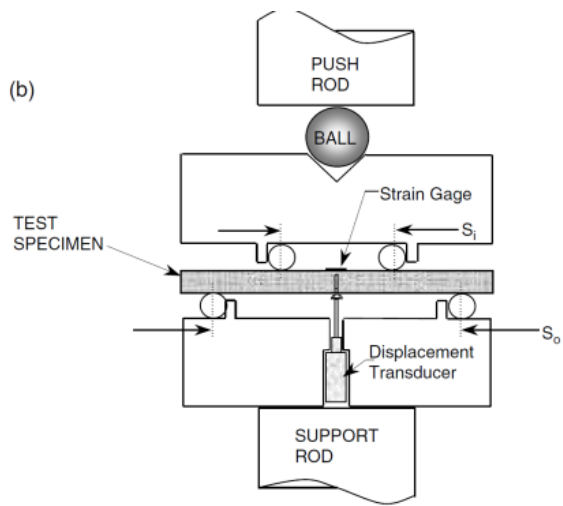
All the methods fulfill the fundamental requirement that a sharp, real crack be used to detect the onset of crack extension. Ideally this extension should be stable, and it generally has been shown to be for both the CNB and SEPB methods.

24

Chapter 4: Thermal and thermo mechanical properties



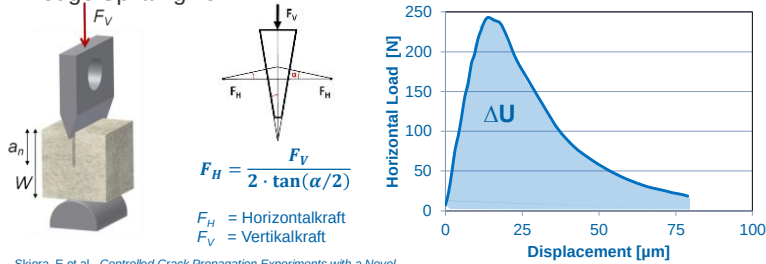
Conference Paper · January 2005
DOI: 10.1007/978-0-387-28920-5_40



Conference Paper · January 2005
DOI: 10.1007/978-0-387-28920-5_40

Chapter 4: Thermal and thermo mechanical properties

Wedge Splitting Test



Skiera, E et al., *Controlled Crack Propagation Experiments with a Novel Alumina-Based Refractory*. Advanced Engineering Materials, 2012.

$$\gamma_{wof} = \frac{\Delta U}{2\Delta A}$$

γ_{wof} = work of fracture
 ΔU = dissipated Energy
 ΔA = Fracture surface

$$R'''' = \frac{E * \gamma_{wof}}{\sigma_f^2}$$

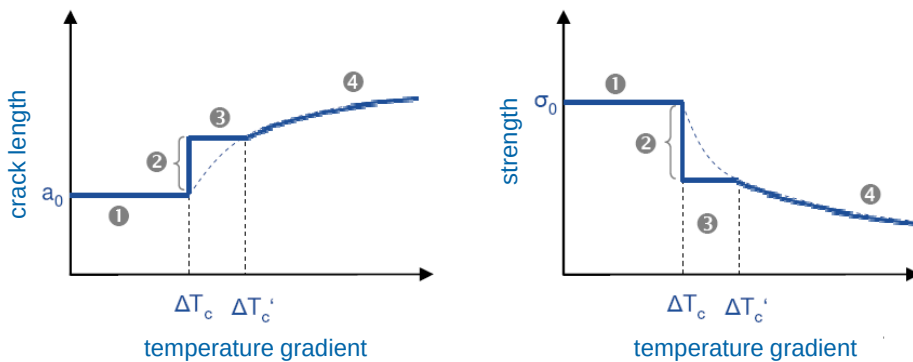
R'''' = Resistance against catastrophic failure
 E = Young's modulus
 σ_f = Fracture strength

Hasselmann, D.P.H., *Thermal Stress Resistance Parameters for Brittle Refractory Ceramics - a Compendium*. American Ceramic Society Bulletin, 1970.

Nano- and micrometre additions of SiO_2 , ZrO_2 and TiO_2 in fine grained alumina refractory ceramics for improved thermal shock performance, Dudczig, S., Veres, D., Aneziris, C.G., Skiera, E., Steinbrech, R.W., *Ceramics International*, 2012, 38(3), pp. 2011-2019

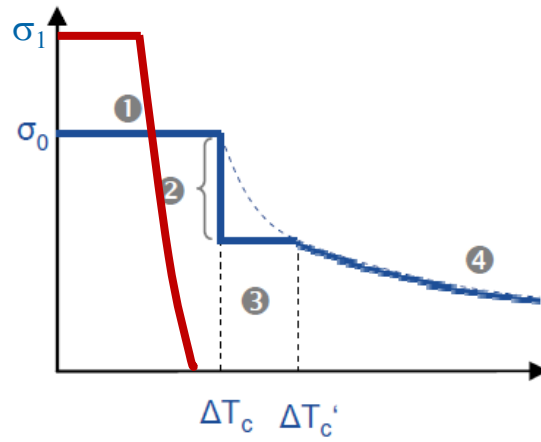
27

Chapter 4: Thermal and thermo mechanical properties



28

Chapter 4: Thermal and thermo mechanical properties



29

Chapter 4: Thermal and thermo mechanical properties

		Al_2TiO_5	SiO_2	Al_2O_3	Si_3N_4	SiSiC
σ	[MPa]	80	100	350	660	330
m		20	20	10	15	13
E	[GPa]	30	90	380	300	360
$\alpha_{20-1000^\circ\text{C}}$	$10^{-6}/\text{K}$	1,5	0,5	9	3	3,2
$\lambda_{1000^\circ\text{C}}$	W/mK	0,5	0,5	2	35	140
ρ	g/cm^3	3,15	2,2	3,99	3,31	3,06
R1	K	1240	1550	70	510	200
R2	W/m	620	775	140	17850	28000

30

Chapter 4: Thermal and thermo mechanical properties

Improved Thermal Stress Resistance Parameters Considering Temperature Gradients for Bricks in Refractory Linings

Erwan Brochen*

Forschungsgemeinschaft Feuerfest e.V. (FGF), 53113 Bonn, Germany

Jürgen Pötschke

45219 Essen, Germany

Christos G. Aneziris

Institute of Ceramic, Glass and Construction Materials, Technische Universität Bergakademie Freiberg,
09596 Freiberg, Germany

31

Chapter 4: Thermal and thermo mechanical properties

Analytical expression	Dimension	Literature designation	Meaning
$R = \frac{\sigma_e \cdot (1 - \nu)}{E \cdot \alpha}$	[K]	R	Thermal stress fracture resistance parameter – maximum allowable temperature difference ΔT_c applied to a prismatic body (severe thermal shock) [3]
$R' = \frac{\sigma_e \cdot (1 - \nu) \cdot \lambda}{E \cdot \alpha}$	[W·m ⁻¹]	R'	Thermal stress fracture resistance parameter – maximum allowable heat flux (mild thermal shock) [4]
$R'' = \frac{\sigma_e \cdot (1 - \nu) \cdot a}{E \cdot \alpha}$	[K·m ² ·s ⁻¹]	R''	Thermal stress fracture resistance parameter – maximum allowable rate of surface heating [4]
$R''' = \frac{G_f \cdot E}{\sigma_f^2 \cdot (1 - \nu)}$	[m]	R'''	Thermal stress damage resistance parameter – damage resistance on initiation of thermal stress fracture [5]
$R_{st} = \sqrt{\frac{G_f}{\alpha^2 \cdot E_e \cdot (1 - \nu)}}$	[K·m ^{1/2}]	R _{st}	Thermal stress damage resistance parameter – damage resistance against quasi-static crack propagation [5]

Where σ_e [MPa] is the strength limit of the material, ν [-] the Poisson's ratio, α [K⁻¹] the thermal expansion coefficient, E [MPa] the modulus of elasticity, a [m²·s⁻¹] the thermal diffusivity and G_f [J·m⁻²] the specific fracture energy

32

Chapter 4: Thermal and thermo mechanical properties

Table II. Estimated Heat Transfer Coefficients From Laboratory Tests

Thermal shock condition	Hot calm air 1000°C	Gas burner ~ 1300°C	Contact with a hot SiC-plate (°C)			Liquid pig iron 1500°C
			500	1000	1500	
h [W/m ² /K]	~ 60	~ 150	~ 250	~ 300	~ 350	>1000

33

Chapter 4: Thermal and thermo mechanical properties

Table III. Guideline Values for Mechanical and Thermal Material Properties of Common Industrial Refractories^{10,21,22}

Materials	σ_c^* [MPa]	$E \cdot 10^{-3}$ [MPa]	G_f [J/m ²]	$\alpha \cdot 10^5$ [1/K]	$\lambda^\dagger$ [W/m/K]	C_p [J/kg/K]	ρ [kg/m ³]	v [—]	Apparent porosity [Vol%]
Corundum bricks >98% Al ₂ O ₃	150	35	200	0.8	6	1050	3300	0.22	20
Magnesia bricks -97% MgO	80	40	200	1.35	15	1300	3100	0.22	12–18
Magnesia-chromite bricks -33% Cr ₂ O ₃	50	25	250	0.9	6	1000	3100	0.22	15–20
Dolomite bricks -60% CaO	60	30	275	1.2	3	1050	2900	0.22	15
Fireclay bricks (A30) 30–35% Al ₂ O ₃	40	15	150	0.6	2	1100	2100	0.22	15–20
Bauxite bricks (B80) 75–85% Al ₂ O ₃	80	35	300	0.7	3	1050	2800	0.22	20

*As no reliable values for the shear strength of refractory materials are available, the resistance to compressive stress was used as characteristic strength feature. In fact, refractory materials under compression load usually fail because of the resulting shearing stresses.

[†]Thermal conductivity at approximately 500°C.

34

Chapter 4: Thermal and thermo mechanical properties

Table IV. Thermal Stress Resistance Parameters of Common Industrial Refractory Products
(Bricks Length L = 0.25 m)

Materials	R [K] ($b = 300 \text{ W/m}^2/\text{K}$)		$\dot{T}_c^*$ [K/s] ($T_{\text{end}} = 1500 \text{ }^\circ\text{C}$)		$R''' \cdot 10^3 \text{ [m]}$ ($\Delta T = 1500 \text{ K}$, $b = 300 \text{ W/m}^2/\text{K}$)		R_{st} [K m ^{1/2}] (Eq. (12))	TSR* (*DIN 51068 or **EN 933-11)
	Kingery (Eq. (2))	Reviewed (GL (21))	Buessem (Eq. (6))	Reviewed (Eq. (24))	Hasselmann (Eq. (9))	Reviewed (Eq. (27))		
Corundum bricks >98% Al_2O_3	418	817	0.035	0.061	0.2	56.8	7.6	Moderate*
Magnesia bricks- 97% MgO	130	357	0.021	0.01	0.8	24.7	4.2	Low**
Magnesia -chromite bricks- 33% Cr_2O_3	173	321	0.016	0.011	1.6	296	8.9	Good**
Dolomite bricks- 60% CaO	130	197	0.006	0.003	1.5	288	6.4	Average**
Fireclay bricks (A30) 30-35% Al_2O_3	347	484	0.014	0.021	0.9	516	13.3	Good*
Bauxite bricks (B80) 75-85% Al_2O_3 B80	254	387	0.012	0.013	1.1	767	10.6	Very good*

*Thermal shock resistance (TSR) with relative evaluation ²².

35

Chapter 4: Thermal and thermo mechanical properties

Symb ol.	Parameter	Comparis on	Environme nt	Failure	Note
R	$\frac{\sigma_c(1-\nu)}{\alpha E}$	ΔT_{max} [K]	Convective for $\beta \gg 1$		
R'	$\frac{\sigma_c(1-\nu)\lambda}{\alpha E}$	$\dot{q}_{\text{max}}$ $\left[\frac{\text{W}}{\text{m}}\right]$	Convective for $\beta < 1$		
R _{cr}	$\frac{\sigma_c(1-\nu)}{\alpha \eta}$	$\Delta \dot{T}_{\text{max}}$ $\left[\frac{\text{K}}{\text{s}}\right]$	Convective for $\beta \gg 1$		
R' _{cr}	$\frac{\sigma_c(1-\nu)\lambda}{\alpha \eta}$	$\ddot{q}_{\text{max}}$ $\left[\frac{\text{W}}{\text{ms}}\right]$	Convective for $\beta < 1$		
R''	$\frac{\sigma_c(1-\nu)a}{\alpha E}$	$(\dot{T})_{\text{max}}$ $\left[\frac{\text{K}}{\text{s}}\right]$	Heating of surface		
R _{rad}	$\left\{ \frac{\sigma_c(1-\nu)\lambda}{\alpha E \varepsilon} \right\}^{1/4}$	T_{max} [K] (blackbody)	Radiation $T_0 \ll T_{\text{max}}$		
R _{transp}	$\left\{ \frac{\sigma_c(1-\nu)\lambda}{\alpha E \varepsilon (1-W_{\Lambda 0})} \right\}^{1/4}$	T_{max} [K] (blackbody)	Radiation $T_0 \ll T_{\text{max}}$		

Source: Franken, Diss., 2000, Aneziris, 2000 D.P.H. Hasselman, 1970

36

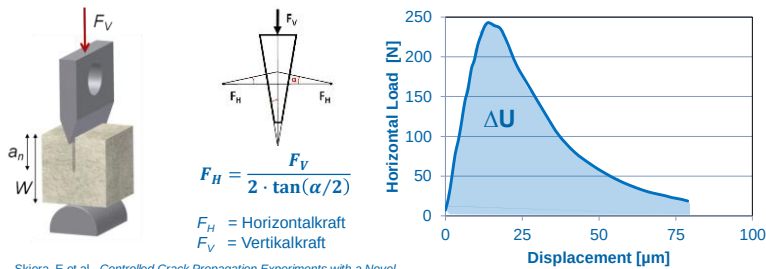
Sym bol	Parameter	Comparis on	Environme nt	Failure	Notes
R^*	$R \left(1 + \frac{(\sqrt{\lambda \rho c_p})_{\text{Probe}}}{(\sqrt{\lambda \rho c_p})_{\text{Medium}}} \right)$	$\Delta T_{\text{max}} [\text{K}]$	β variabel	Crack initiation	Source: Scholz
R'''	$\frac{E}{\sigma_c^2 (1 - \nu)}$	$\frac{1}{W_{\text{el}}} \left[\frac{\text{m}^2}{\text{N}} \right]$	-		min. elast. energy on fracture
R''''	$\frac{G_f E}{\sigma_c^2 (1 - \nu)}$	$[\text{m}]$	-		min. crack growth
T	$-\log \left[\frac{\sigma_c^2}{G_f E} \right]$	$\left[\log \left(\frac{1}{\text{m}} \right) \right]$	-		Resistance to fragmentati on
R_{st}	$\left[\frac{G_f}{\alpha^2 E} \right]^{1/2}$	$\Delta T_{\text{max}} [\text{K}]$	β large		Long cracks
R'_{st}	$\left[\frac{G_f \lambda^2}{\alpha^2 E} \right]^{1/2}$	$\Delta T_{\text{max}} [\text{K}]$	β small		Long cracks

Source: Franken, Diss., 2000, Aneziris, 2000 D.P.H. Hasselman, 1970

37

Chapter 4: Thermal and thermo mechanical properties

Wedge Splitting Test



Skiera, E et al., Controlled Crack Propagation Experiments with a Novel Alumina-Based Refractory, Advanced Engineering Materials, 2012.

$$\gamma_{wof} = \frac{\Delta U}{2\Delta A}$$

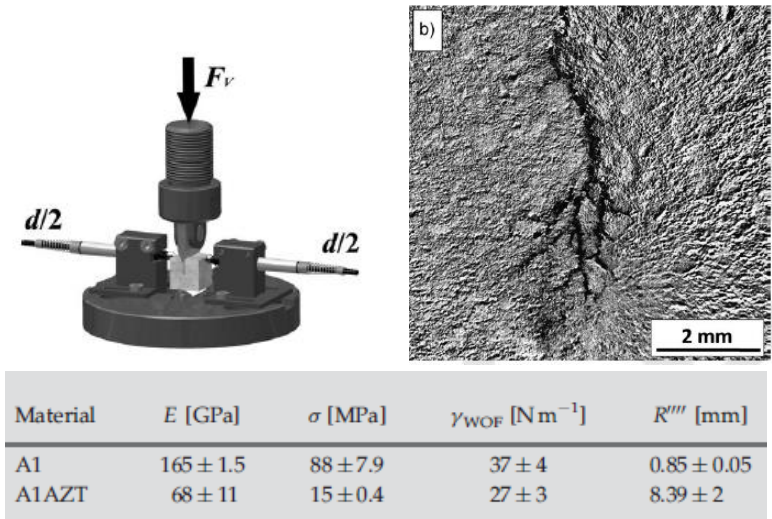
γ_{wof} = work of fracture
 ΔU = dissipated Energy
 ΔA = Fracture surface

$$R'''' = \frac{E * \gamma_{wof}}{\sigma_f^2}$$

R'''' = Resistance against catastrophic failure
 E = Young's modulus
 σ_f = Fracture strength

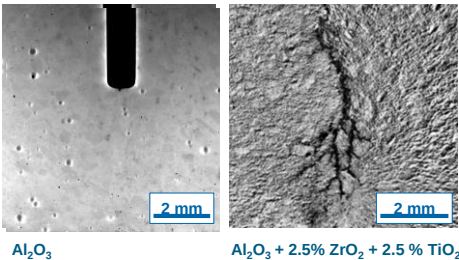
Hasselman, D.P.H., Thermal Stress Resistance Parameters for Brittle Refractory Ceramics - a Compendium. American Ceramic Society Bulletin, 1970.

Chapter 4: Thermal and thermo mechanical properties



DOI: 10.1002/adem.201100221

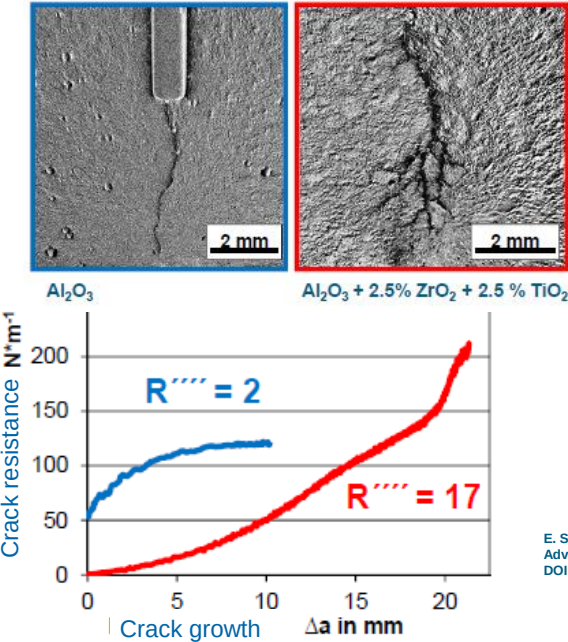
Chapter 4: Thermal and thermo mechanical properties



- Aneziris, C.G. et al., Adv.Eng.Mat. 12 (2010) 6, 478-485;
- Skiera, E. et al., Adv.Eng.Mat. (2011), DOI:10.1002/adem.201100221;
- Dudczig, S. et al., Ceram.Int. (2011), DOI:10.1016/j.ceramint.2011.10.036;
- Dudczig, S. et al., Refractories Worldforum 4 (2012),105-111;
- Berek, H. et al., Refractories Worldforum 4 (2012), 85-90

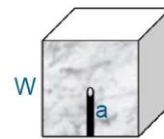
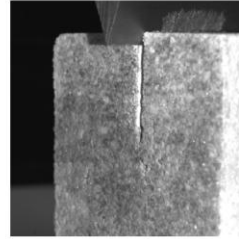
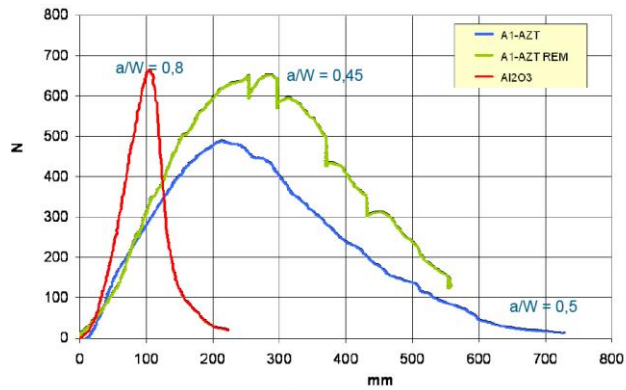
Material	E [GPa]	σ [MPa]	γ_{WOF} [N m^{-1}]	R'''' [mm]
Al	165 ± 1.5	88 ± 7.9	37 ± 4	0.85 ± 0.05
AlAZT	68 ± 11	15 ± 0.4	27 ± 3	8.39 ± 2

E. Skiera, J. Malzbender, C. G. Aneziris et al
Advanced Eng. Mater.
DOI: 10.1002/adem.201100221



E. Skiera, J. Malzbender, C. G. Aneziris et al
Advanced Eng. Mater.
DOI: 10.1002/adem.201100221

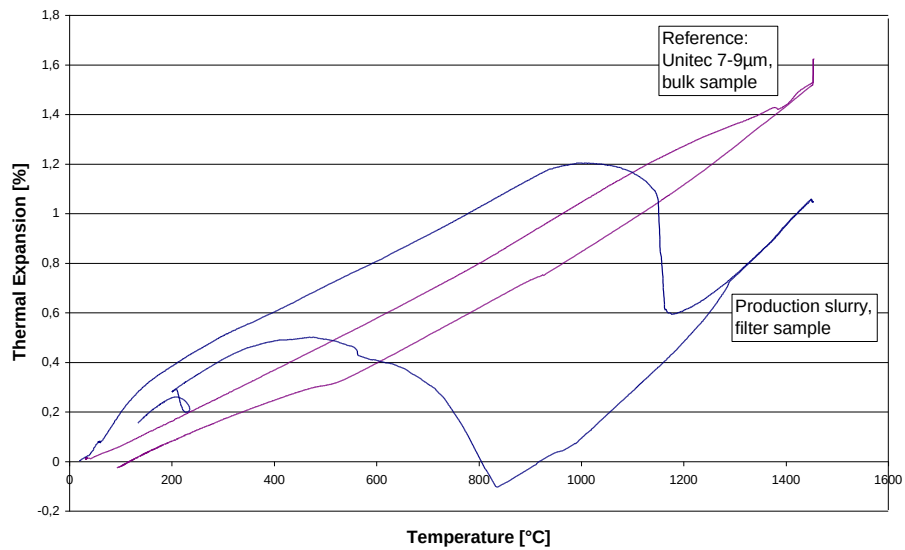
Chapter 4: Thermal and thermo mechanical properties



DOI: 10.1002/adem.201100221

43

Chapter 4: Thermal and thermo mechanical properties



44

Chapter 4: Thermal and thermo mechanical properties

How can we measure:

- Residual strength
- Change of E
- Work of fracture
- Change of weight

Pay attention:

K_{Ic} is not the same as thermal shock performance!

45

Chapter 4: Thermal and thermo mechanical properties

Lienhard, J.H. IV, Lienhard, J.H. V, : A Heat Transfer Textbook. Phlogiston Press, Cambridge, MA, 2016.

Ashby, M.F.: Materials selection in mechanical design. Elsevier Butterworth-Heinemann, Oxford, 2005.

Hasselman, D.P.H.: Unified Theory of Thermal Shock Fracture Initiation and Crack Propagation in Brittle Ceramics. *Journal of the American Ceramic Society* 52 (1969), pp. 600-604.

Kingery, W.D.: Factors Affecting Thermal Stress Resistance of Ceramic Materials. *Journal of the American Ceramic Society* 38 (1955), pp. 3-15.

Brochen, E., Pötschke, J., Aneziris, C.G.: Improved Thermal Stress Resistance Parameters considering Temperature Gradients for Bricks in Refractory Linings. *International Journal of Applied Ceramic Technology* 11 (2014), pp. 371-383.

Skiera, E.: Thermomechanische Charakterisierung neu entwickelter Feuerfestwerkstoffe. Dissertation, RTWH Aachen, 2012.

Brochen, E.: Messung und Modellierung der Temperaturwechselbeständigkeit feuerfester Werkstoffe. Dissertation, TU Bergakademie Freiberg, 2011.

46